

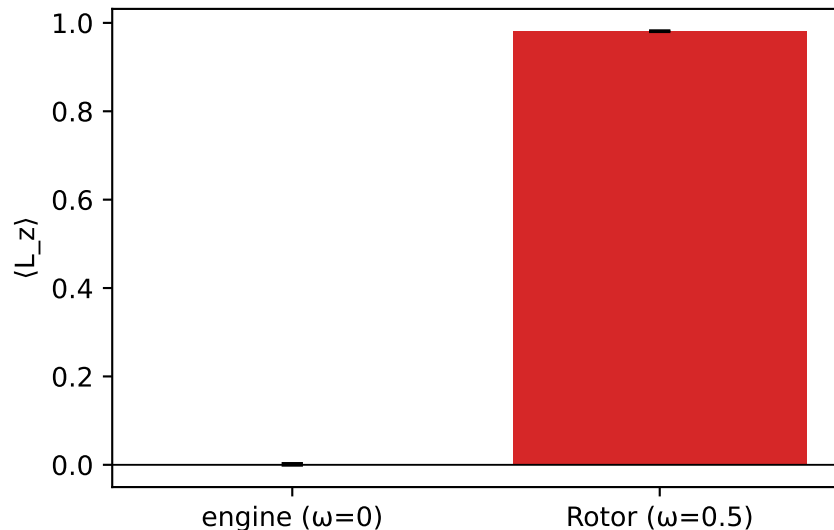
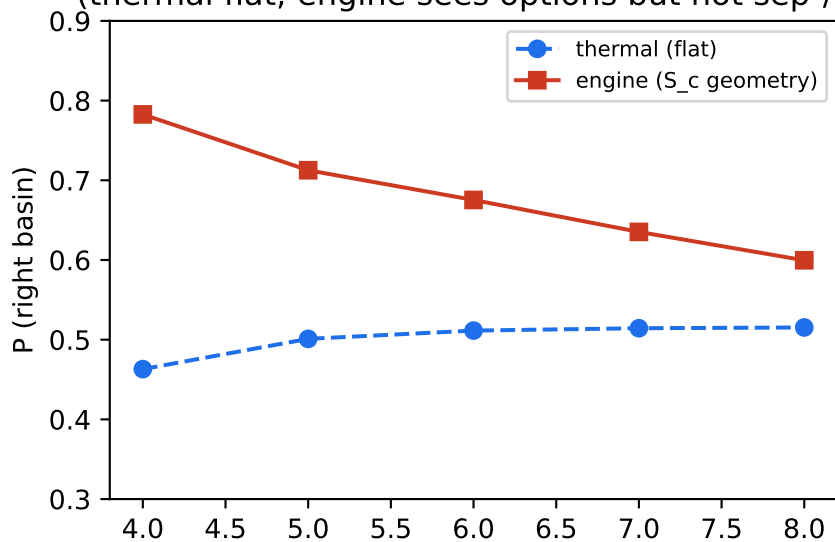
Fig 7 — Arc (honest): v3 re-confirms v2 thermodynamic equivalence in conservative 1D.

The negative (no unique τ -fingerprint beyond U_{eff}) is the result. Frontier = v4 non-conservative. Data: manifest (8000 traj/cell).

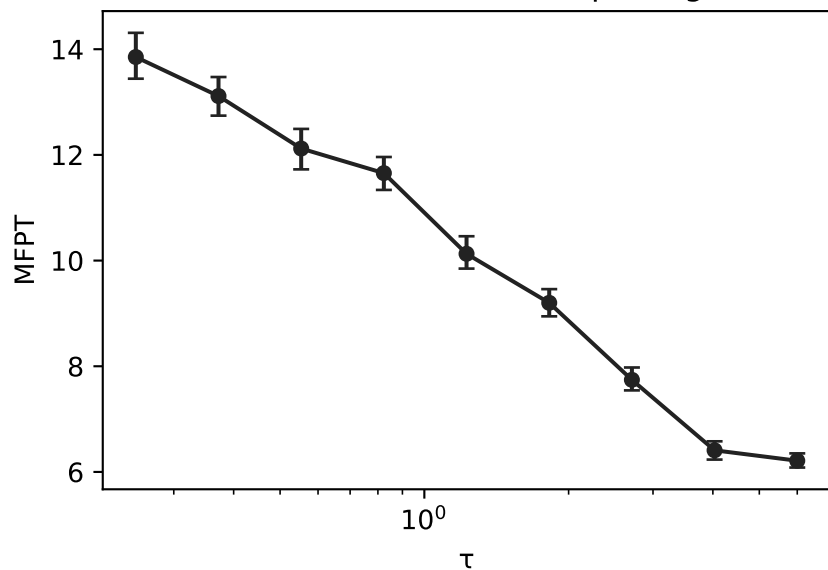
v2: current test — engine = 0 (conservative)

positive control detects real NESS
(this is the measurement)

v1/v2: occupancy blind to L (detailed balance)
(thermal flat; engine sees options but not sep^2/D)



v3: MFPT(τ) dial^{sep} — monotone, kinetic
(the horizon moves first-passage)



v3 honest negative (this run)

- τ produces a visible FPT dial (real).
- But the dial *is* equilibrium relaxation in the τ -dependent $U_{\text{eff}}(\tau)$ — by definition once Arm A (conservative) is granted.
- $\alpha \approx 1$ (drift under force), not thermal diffusive $\alpha=2$.
- Zero current re-confirms thermodynamic equivalence (fig 5).

Conclusion: in conservative 1D the τ -knob does not produce a fingerprint no static U can mimic. The frontier is v4 (non-conservative force, 2D+, curl $\neq 0$).